Aaron E. Wu

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Education

Georgia Institute of Technology | Atlanta, GA *Fall 2021 - Present*

Ph.D. in Robotics (GPA 4.00, expected Spring 2029) | **B.S. in Electrical Engineering** (GPA 4.00, 2024)

Relevant Coursework: Deep Reinforcement Learning, Computer Vision, Networked Control, Advanced Mobile Robotics

Awards: PURA Salary, Faculty Honors, Eta Kappa Nu Outstanding Senior Scholar, ECE Senior Scholar

Skills

ML/Edge Compute: Reinforcement Learning, Sim2Real Transfer, YOLO, NanoSAM, CUDA, TensorRT

Software & Infrastructure: Python, C/C++, ROS 2 (Humble/Jazzy), Linux, Docker, Colcon, Git, Bash, Protobuf

Hardware: Jetson Orin Nano, Stereo RGB-D cameras, Embedded HITL Systems, Electrical System Design

Experience

Aerospace Systems Design Lab | Dr. Dimitri Mavris | Atlanta, GA *Fall 2024 – Present*

ONR-funded research building the Unmanned Collaborative Research Testbed (UCRT), a multi-domain robotic testbed.

- Built Gazebo-based simulator generating 100k+ trajectories for training residual models for 3-DOF marine dynamics; set up simulation computers and implemented a custom package installation system to streamline software deployment
- Developed ROS 2 navigation and sensor fusion pipelines for multi-domain robotic testbed; containerized systems using Docker for deployment across air, ground, and marine platforms
- Designed Hardware-in-the-Loop testing framework for autonomous surface vehicles, enabling rapid validation of localization and state estimation algorithms in real-time embedded environments
- Founded and taught AE x355, a hands-on marine robotics design course, onboarding new members through the design, integration, and field testing of the *Stinger Tug* autonomous surface vehicle.
- Mentored undergraduate and graduate researchers across software, hardware, and systems engineering, helping grow MRG into a research-driven organization aligned with ONR-funded initiatives.

RF Engineering Intern | SpaceX | Seattle, WA *Spring 2024 – Fall 2024*

Summer internship to build up new antenna measurement systems for phased arrays at SpaceX within a strict launch deadline.

- Developed python automation software for phased array calibration systems, reducing test time from 3 hours to 30 seconds and enabling production scaling for next-generation satellite arrays
- Collaborated with system validation RF and software engineering team to optimize system performance strict deadlines

Undergraduate Research | Propagation Group| Dr. Gregory Durgin | Atlanta, GA *Spring 2023 – Fall 2023*

Individual research into perfect pulse modulation scheme to allow for more power efficient RF propagation in seawater. Research stemmed from Dr. Durgin's ECE 4390 (Radar and EM Sensing) class.

- Analyzed transfer function between two antennas underwater in HFSS to propose novel, efficient modulation scheme
- Developed novel modulation scheme for underwater RF communication by analyzing transfer functions in HFSS; awarded President's Undergraduate Research Award ($1500); won best poster at IEEE DTPI 2023

Undergraduate Intern | Scripps Institution of Oceanography MPL REU | Dr. Florian Meyer | San Diego, CA *Summer 2023*

Undergraduate research experience working with PI and graduate students to build up the lab's robotic testing infrastructure, primarily involving an autonomous surface vehicle (ASV).

- Designed and integrated side-scan sonar navigation system for ASV; conducted ocean field tests generating datasets for novel localization algorithm validation

Undergraduate Research | mmWave Antennas & Arrays Lab | Dr. Nima Ghalichechian | Atlanta, GA *Fall 2022 – Spring 2023*

Project to create robot arm capable of measuring antenna radiation pattern from 6-110 GHz by employing difficult near to far field transformations. Worked closely with Ph.D. student and other undergraduates to take ownership of the project.

- Designed and fabricated custom antenna mounts for pattern measurement system; mounts remain in active use by the lab today as core infrastructure for ongoing antenna research

Publications

- Ortiz Solano, J. L., et al. (incl. Wu, A.) "Systems Engineering Multi-Domain Robotic Field Testing," IEEE OCEANS 2025
- Devlin, T., et al. (incl. Wu, A.) "Reconfigurable USV Testbed for Multi-Agent Field Robotics," IEEE OCEANS 2025
- Chan, A., et al. (incl. Wu, A.) "Stinger Tug: Educational Robots in the Maritime Domain," IEEE OCEANS 2025
- Wu, A. E., et al. "Perfect Pulses to Improve Acoustic and RF Underwater Wireless Communication," IEEE ORSS, 2023
- Meng, M., Wu, A., et al. "Robotic Radiation Pattern Measurement System for 6–110 GHz," IEEE ORSS, 2023